# Anatomy of an Excellent Engineering Prompt

Each component serves a specific purpose in guiding AI response quality

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| **Role Assignment** | "Act as a senior electromechanical engineer specializing in automotive systems" | Sets expertise level and perspective for response |
| **Context** | "I am designing a regenerative braking system for an electric vehicle" | Establishes the problem domain and application |
| **Specifications** | "Vehicle mass: 2000kg, max speed: 150km/h, battery: 400V nominal" | Provides concrete parameters for calculations |
| **Constraints** | "Must comply with ISO 26262 ASIL−B, budget under $500/unit" | Defines boundaries and requirements |
| **Deliverables** | "Provide: 1) Energy calculations, 2) Component selection, 3) Control strategy" | Lists exactly what you need from the AI |
| **Format** | "Format as a technical design document with diagrams described" | Ensures output is immediately usable |
| **Standards** | "Reference ISO 26262 and SAE J2954 where applicable" | Grounds response in industry requirements |